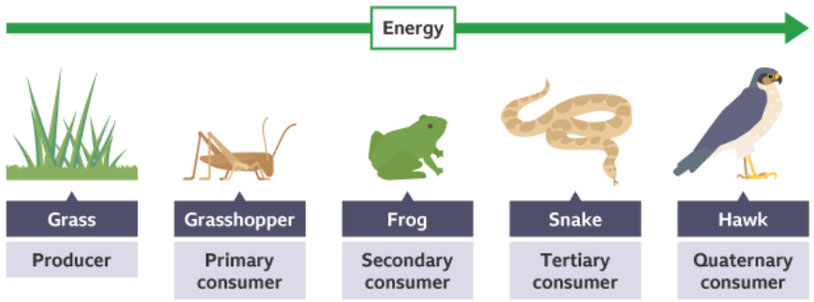
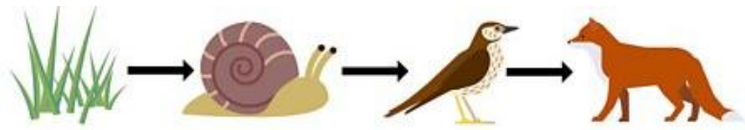
Knowledge Organiser
Year 7: Interdependence – Food Webs and Chains

What does "interdependence" mean?
- All **organisms** depend on other organisms for food and energy in an ecosystem
- A change in the number of organisms in one species will affect the other species in the ecosystem
- The feeding relationships between organisms are represented by **food chains** and **food webs**

What is a food chain?
- A food chain is a diagram showing the **feeding relationships** between organisms
- It shows the **flow of energy**, from left to right
- Organism are classified as either **producers** or **consumers**
- A producer makes its own food e.g. a plant through **photosynthesis**
- A consumer eats other organisms to get energy e.g. a grasshopper eating a plant

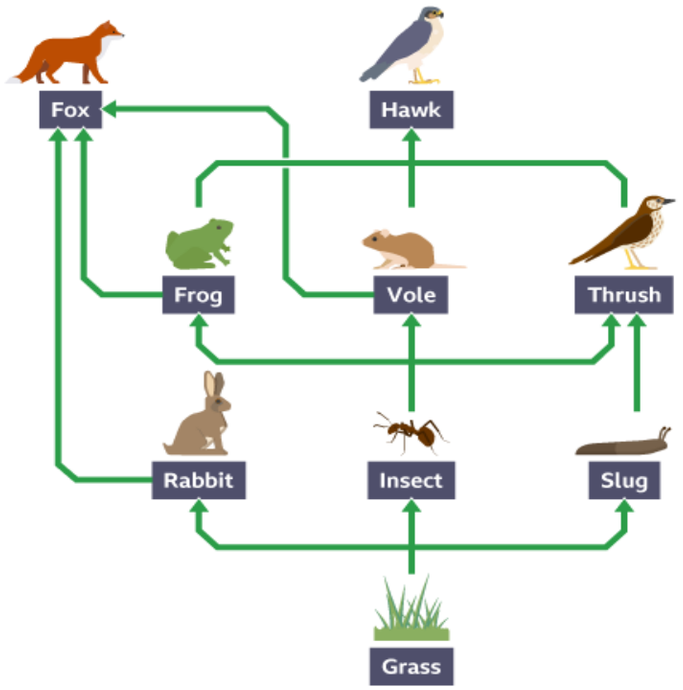


A labelled diagram giving examples of producers and consumers in a food chain



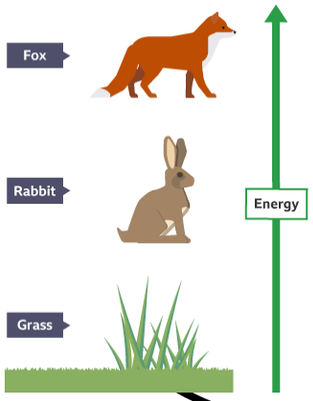
Organism	Trophic level
Grass	1st: Producer
Snail	2nd: Primary consumer
Thrush	3rd: Secondary consumer
Fox	4th: Tertiary consumer

How is a food web different?
- A food web shows how different feeding relationships affect each other
- It is drawn as a **network of connected food chains** in the ecosystem
- Producers are shown at the bottom and the food chains go upwards



This food web has several food chains:
- grass → insect → vole → hawk
- grass → insect → frog → fox
- grass → insect → vole → fox

Some organisms have just one consumer, others like the frog or vole have two.

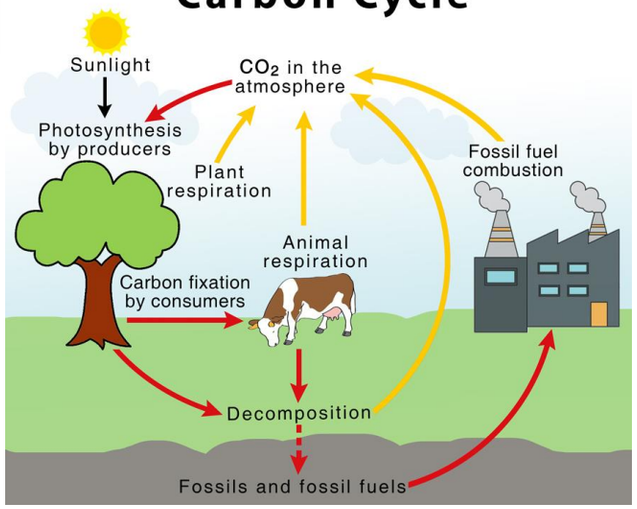


What is a trophic level?
- A **trophic level** indicates an organisms position in a food chain
- Producers are at the "**first trophic level**" because they make the first level of any food chain
- Primary consumers make the **"second trophic level"** because they eat producers

Food Chain Key Points:
1. Each food chain starts with a producer
2. Each level in a food chain is called a "trophic level"
3. The arrow in a food chain represents the flow of energy
4. The amount of energy in a food chain decreases with each trophic level

What are decomposers?
- **Decomposers** feed on material from dead organisms
- They play an important role in an ecosystem by breaking up their remains into **simpler materials**
- These materials can then be re-used by other organisms e.g. nitrogen by plants
- Examples of decomposers include earthworms, bacteria and fungi

Carbon Cycle



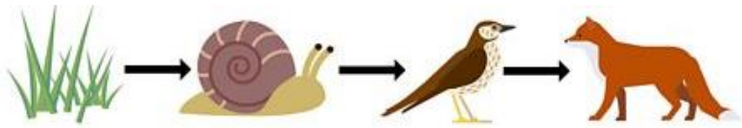
A diagram of the **carbon cycle**.

It shows how carbon is returned to the atmosphere from the decomposition of plants and animals.

RAG Yourself!

Learning Outcomes;	RAG
Draw food chains	
Draw food webs	
Describe what is meant by interdependence	
Predict the effect of an environmental change on a food web	
Describe the role of decomposers in an ecosystem	
Use the words "producers" and "consumers"	

Knowledge Organiser
Year 7: Interdependence – Food Webs and Chains



What does “interdependence” mean?

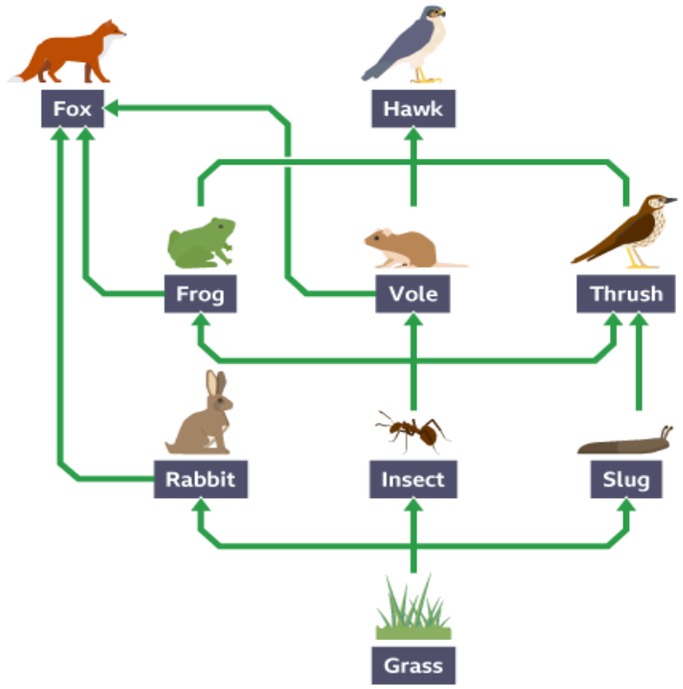
Organism	Trophic level
Grass	
Snail	
Thrush	
Fox	

Food Chain Key Points:

What is a food chain?

How is a food web different?

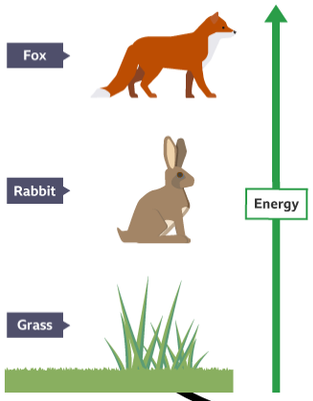
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A diagram of the carbon cycle.

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A labelled diagram giving examples of producers and consumers in a food chain



What is a trophic level?

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